

6 Some processes in plants and animals are under the control of chemicals.

(a) (i) Which row in the table shows the effect of an increase in adrenaline in a mammal?

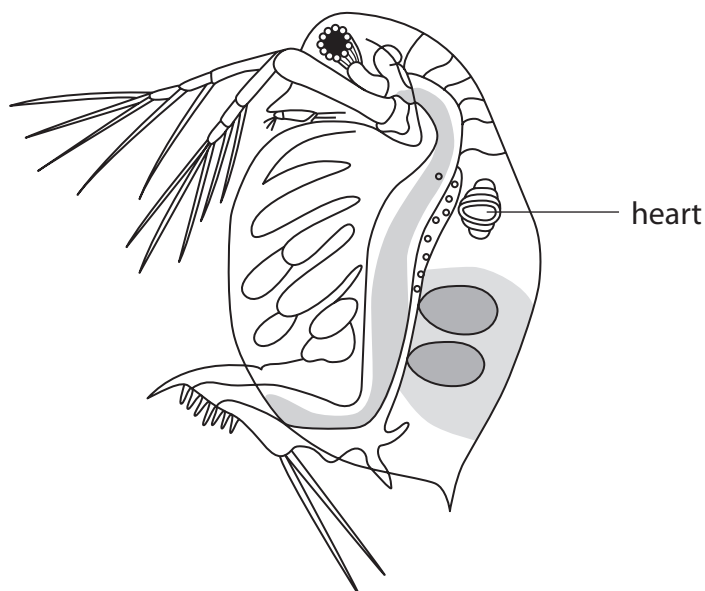
(1)

		breathing rate	heart rate
☐	A	decreases	increases
☐	B	increases	decreases
☐	C	increases	increases
☐	D	decreases	decreases

(ii) Nicotine is a drug found in tobacco products.

The effect of changing the concentration of nicotine and adrenaline on the mean heart rate of *Daphnia* was investigated.

The diagram shows a *Daphnia*.



The results are shown in the table.

Concentration of nicotine / mmol dm^{-3}	Mean heart rate / bpm		
	No adrenaline	Concentration of adrenaline $0.001 \text{ mmol dm}^{-3}$	Adrenaline 0.1 mmol dm^{-3}
1000	84	81	104
10	96	99	110
0.1	103	105	113
0.01	108	109	127
0.001	106	107	151
0.00 (control)	146	160	

Deduce the effect of the combination of these chemicals on the heart rate of *Daphnia*.

(3)

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(b) Which is a description of a gibberellin?

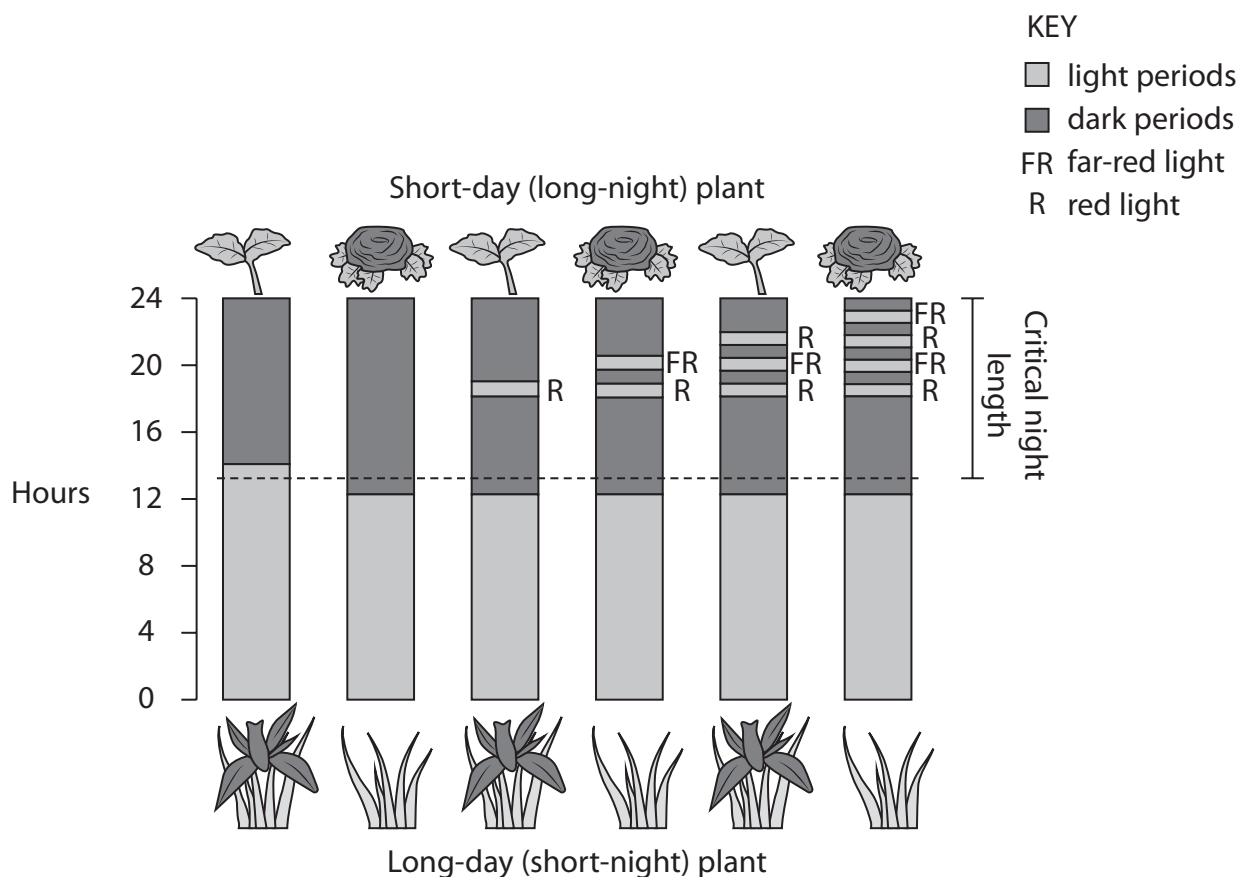
(1)

- ☐ **A** an enzyme that converts starch to glucose
- ☐ **B** a molecule that converts phytochrome far-red (Pfr) to phytochrome red (Pr)
- ☐ **C** an organic compound that stimulates the action of transcription factors
- ☐ **D** a protein that wraps around DNA

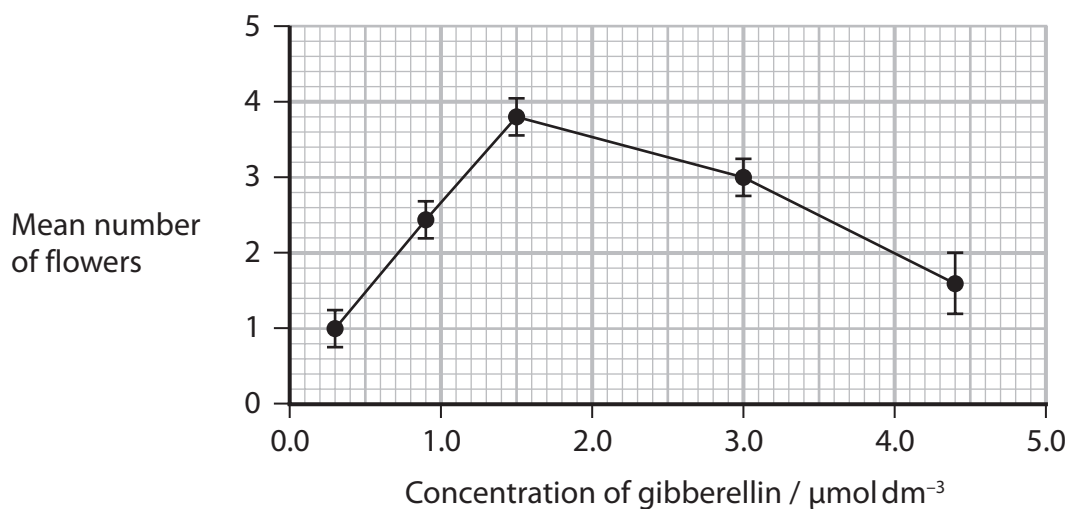
*(c) The quality and length of daylight act as signals to initiate and regulate flowering in plants. This involves phytochromes and the production of gibberellins.

In an investigation, a long-day plant and a short-day plant were exposed to light for different lengths of time. During dark periods they were exposed to flashes of red and far-red light.

The diagram shows the results of this investigation.



The graph shows the results of an investigation into the effect of gibberellin concentration on the flowering of another type of plant.



Use your own knowledge and the information in the question to support your answer.

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(Total for Question 6 = 11 marks)